

GRAIN AND TILLER: LOSS-INFORMED RATIONAL TRANSFORMER UPDATES

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ABSTRACT

1 INTRODUCTION

2 RELATED WORK

3 METHOD

A learned activation changes both the features a network represents and how parameter updates affect those features. We use this connection to design GRAIN and TILLER together. GRAIN learns a rational response within each feature group. TILLER uses the resulting response geometry to construct update directions, then coordinates these directions through their joint effect on the loss. This separates two decisions: which direction to take within each group, and how to combine directions across the network.

3.1 GRAIN: LEARNING THE RESPONSE WITHIN EACH GROUP

Consider a feed-forward block with input $x \in \mathbb{R}^d$. Partition its intermediate features into G groups of width w , with incoming matrices $A_g \in \mathbb{R}^{w \times d}$ and outgoing matrices $B_g \in \mathbb{R}^{d \times w}$. The block computes

$$y = \sum_{g=1}^G B_g h_g, \quad h_g = \rho_g \phi_g(u_g), \quad u_g = \frac{A_g x}{\rho_g}, \quad \rho_g = \sqrt{\|A_g x\|_2^2 / w + \epsilon_{\text{act}}}. \quad (1)$$

Each ϕ_g is a learned rational function, applied elementwise:

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

Rational functions offer a trainable response shape (Molina et al., 2020; Boullé et al., 2020); the denominator above stays positive. Internal RMS normalization (Zhang & Sennrich, 2019) lets the function act on relative feature magnitudes, while multiplication by ρ_g restores the group scale. Groups start from a common SiLU fit and learn separate responses. With intermediate width $Gw = 3H/2$, the two projections use the same number of dense weights as a SwiGLU block of width H (Shazeer, 2020).

This parameterization also exposes the response of a group to an update. Writing $J_g = \partial h_g / \partial (A_g x)$, differentiation gives

$$J_g = \text{Diag}(\phi'_g(u_g)) + \frac{1}{w} [\phi_g(u_g) - u_g \odot \phi'_g(u_g)] u_g^\top. \quad (3)$$

The first term describes the elementwise response; the second captures the coupling induced by normalization. Both learned features h_g and their Jacobian J_g inform the optimizer below. Appendix A derives this identity and specifies initialization.

3.2 TILLER: RESPONSE-GUIDED DIRECTIONS AND LOSS-AWARE COORDINATION

For L feed-forward layers, index the $C = LG$ layer-group pairs by $i = (\ell, g)$. Write g_i for a block’s current clipped loss gradient. For each of its two matrix roles, start with a Muon direction p_i and a direction a_i obtained from factorized second moments (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). Their orthogonal component $a_i^\perp = a_i - \langle a_i, p_i \rangle p_i / \|p_i\|_F^2$ provides a way to change the direction while controlling its norm.

To determine the rotation magnitude, TILLER measures two properties of the learned response. Participation π_i measures the singular-value energy of J_i for an incoming matrix, or the coordinate energy of h_i for an outgoing matrix. Congruence κ_i compares the live and initial response kernels, using $J_i J_i^\top$ for the incoming role and $h_i h_i^\top$ for the outgoing role. Both statistics are weighted by downstream loss sensitivity. Their definitions and bounds appear in Appendix B. With $e_i = \pi_i(1 - \kappa_i^2) \in [0, 1]$, the candidate direction is

$$\tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp, \quad \sigma_i = \text{sign} \langle g_i, a_i^\perp \rangle. \quad (4)$$

We choose $\sigma_i = 1$ when the contraction is zero. Thus the fixed direction norm moves toward the adaptive tangent only when the response is both broadly supported and geometrically distinct from its initialization. Orthogonality gives $\|\tilde{q}_i\|_F = \|p_i\|_F$: the response changes direction without changing its magnitude. The selected q_i is this candidate when both it and the parent have positive gradient contraction; degenerate or nonfinite candidates retain p_i .

Directions in separate parameter blocks can affect the same loss simultaneously. TILLER measures this relationship in a shared set of activation probes. For probe k and group i , let $\xi_{k\ell}$ be the downstream loss cotangent. A paired direction (q_i^A, q_i^B) contributes

$$S_{ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle. \quad (5)$$

Thus each column describes a group direction, and each row compares all groups at the same probe. For coefficients c , Sc is their combined linearized response. The cross terms in $\|Sc\|_2^2$ distinguish directions whose effects reinforce one another from those whose effects offset.

TILLER maintains a discounted history of these score rows using a Robust Frequent-Directions sketch (Ghashami et al., 2016; Luo et al., 2019). It retains dominant combinations of loss responses and assigns an isotropic remainder to the discarded spectrum. Between probe refreshes, the current minibatch contractions $s_i = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$ supply fresh directional information. This combines the current descent signal with recent information about how directions act together. Appendices C and D give the updates.

Let G_t denote the second-moment metric of loss responses used for the current step and r_t the measured interaction with decoupled weight decay. With scheduled learning rate η , TILLER chooses signed coefficients through the model

$$\min_c -\eta s_t^\top c + \frac{\eta^2}{2} (c^\top G_t c + 2r_t^\top c), \quad \sum_i \omega_i c_i^2 = \sum_i \omega_i, \quad \omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2. \quad (6)$$

The linear term rewards descent on the current batch. The response metric measures squared linearized loss responses across probes; the constraint fixes their total update energy. Coefficients can amplify, attenuate, or reverse individual groups. This makes coordination a choice of how to allocate a shared update norm according to loss information.

A projected solver proposes coefficients; the proposal is accepted when it improves this model, meets the norm constraint, and passes the layerwise descent checks in Appendix E. Otherwise $c_i = 1$. With weight decay λ_{wd} , each paired matrix updates as $A_i \leftarrow (1 - \eta \lambda_{\text{wd}}) A_i - \eta c_i q_i^A$ and analogously for B_i . Attention uses a response-guided rotation based on the same layer’s group statistics; rational coefficients and the remaining parameters use AdamW (Loshchilov & Hutter, 2019). The full update is in Appendix F.2.

4 EXPERIMENTS

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with code implementation, mathematical cross-checking, literature retrieval, and manuscript drafting. The authors reviewed the resulting material and take responsibility for the final content.

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A GRAIN PARAMETERIZATION

GRAIN separates the shape of an activation from the scale of its input group. The rational function learns the shape in normalized coordinates; the group RMS carries the input scale through the block. This construction also describes how a matrix update changes the features.

A.1 FORWARD MAP

Let $H_r = Gw$ be the expanded width, with $w = 256$. For an input $x \in \mathbb{R}^d$, one block is

$$z = Ax, \quad h = \Phi(z), \quad y = Bh, \quad (7)$$

where $A \in \mathbb{R}^{H_r \times d}$ and $B \in \mathbb{R}^{d \times H_r}$. Partition z into G contiguous groups $z_g \in \mathbb{R}^w$. At layer ℓ , write $A_{\ell g} \in \mathbb{R}^{w \times d}$ for the corresponding row block of A_ℓ and $B_{\ell g} \in \mathbb{R}^{d \times w}$ for the corresponding column block of B_ℓ . Then

$$z_{\ell g} = A_{\ell g} x_\ell, \quad y_\ell = \sum_{g=1}^G B_{\ell g} \Phi_g(z_{\ell g}). \quad (8)$$

For each group, define

$$\rho_g = \sqrt{w^{-1} \|z_g\|_2^2 + \epsilon_{\text{act}}}, \quad u_g = z_g / \rho_g, \quad \epsilon_{\text{act}} = 10^{-6}. \quad (9)$$

Each group has a learned rational response with a degree-five numerator and degree-four denominator,

$$P_g(u) = \sum_{k=0}^5 a_{gk} u^k, \quad (10)$$

$$Q_g(u) = 1 + \sum_{k=1}^4 |b_{gk}| |u|^k, \quad (11)$$

$$\Phi_g(z_g) = \rho_g \phi_g(u_g), \quad \phi_g(u) = P_g(u) / Q_g(u), \quad (12)$$

where all functions and powers act elementwise. This combines learned rational activations (Molina et al., 2020; Boullé et al., 2020) with an internal RMS coordinate system (Zhang & Sennrich, 2019). Because $Q_g(u) \geq 1$, the response is finite for every real input.

For a SwiGLU intermediate width H , we set $H_r = 3H/2$. The two dense matrices in equation 7 then contain $2dH_r = 3dH$ weights, equal to the dense projection count of the three SwiGLU matrices. The activation adds $10G$ rational coefficients per block. The learned response supplies the Jacobian and feature statistics used below.

A.2 HOW AN INPUT PERTURBATION CHANGES THE RESPONSE

An incoming matrix direction changes z_g , and hence changes both the normalized features and their RMS scale. Differentiating these two paths reveals their separate contributions to the response.

The normalized map has a diagonal-plus-rank-one Jacobian wherever the rational response is differentiable. Write $f_g = \phi_g(u_g)$, $d_g = \phi'_g(u_g)$, and $r_g = f_g - u_g \odot d_g$, all in $\mathbb{R}^w$. Direct differentiation gives

$$J_g \equiv \frac{\partial \Phi_g}{\partial z_g} = \text{Diag}(d_g) + \frac{1}{w} r_g u_g^\top, \quad J_g^\top v = d_g \odot v + u_g \text{ mean}(r_g \odot v). \quad (13)$$

The diagonal term acts on each feature separately. The rank-one correction couples features through the change in group scale. For an input perturbation v , $J_g v$ is its first-order feature change; for a downstream cotangent v , $J_g^\top v$ carries loss sensitivity back to the group input. These are the two operations used to construct the loss measurements in Appendix C.

Proposition 1 (Pole freedom and Jacobian structure). *For every finite group input, $Q_g(u) \geq 1$. Wherever the absolute-value terms are differentiable, the Jacobian of equations 9 and 12 is equation 13. At $u_j = 0$, where the response need not be classically differentiable, we use the automatic-differentiation selection $\text{sign}(0) = 0$; throughout the paper, J_g denotes this selected derivative at such coordinates.*

Proof. Every term $|b_{gk}| |u|^k$ is nonnegative, hence $Q_g(u) \geq 1$. Suppress the group index and write $\rho = (w^{-1} z^\top z + \epsilon_{\text{act}})^{1/2}$ and $u = z/\rho$. Then

$$\nabla_z \rho = u/w, \quad \frac{\partial u}{\partial z} = \rho^{-1} (I - uu^\top/w). \quad (14)$$

Applying the product and chain rules to $\Phi(z) = \rho\phi(u)$ gives

$$J = \phi(u)u^\top/w + \text{Diag}(\phi'(u)) (I - uu^\top/w) \quad (15)$$

$$= \text{Diag}(d) + (\phi(u) - u \odot d) u^\top/w. \quad (16)$$

Transposition and multiplication by v yield the pullback in equation 13. $\square$

A.3 INITIALIZATION

All groups begin from the same SiLU fit. In ascending powers, the coefficients are

$$(a_0^{\text{init}}, \dots, a_5^{\text{init}}) = (2.4915714620520875 \times 10^{-5}, 0.5000668663485823, 0.2499441908995771, \\ 0.0526200910151016, 0.005525765640115002, 0.00024199321178747251), \quad (17)$$

$$(b_1^{\text{init}}, \dots, b_4^{\text{init}}) = (0.00022554017910995938, 0.10511420350691994, \\ 2.8656992394109812 \times 10^{-5}, 0.0004816962110562025). \quad (18)$$

The ten coefficients subsequently learn independently in every group.

B RESPONSE-AWARE STRUCTURED DIRECTIONS

TILLER combines two views of a matrix update. The Muon direction uses the singular-vector geometry of momentum, while factorized adaptation rescales momentum using the history of gradient magnitudes. The learned activation determines how far the Muon direction rotates in the plane spanned by the two directions. The construction below keeps the Muon norm and uses loss sensitivity to orient the rotation.

B.1 TWO DIRECTIONS FROM THE SAME GRADIENT

Let g_t denote the parameter gradient after microbatch accumulation, distributed averaging, and global-norm clipping. For each pair of group blocks $i = (\ell, g)$ and role $r \in \{A, B\}$, TILLER starts from a Nesterov–Muon direction $p_{t,i}^r$: with $\beta_1 = 0.95$, it forms

$$v_{t,i}^r = \beta_1 v_{t-1,i}^r + (1 - \beta_1) g_{t,i}^r, \quad m_{t,i}^r = \beta_1 v_{t,i}^r + (1 - \beta_1) g_{t,i}^r, \quad (19)$$

followed by a five-step Newton–Schulz polar map applied independently to each group block and Muon’s role-dependent RMS adjustment (Jordan et al., 2024; Bernstein & Newhouse, 2024; Liu et al., 2025). The sum of the nominal block ranks is $G \min\{w, d\}$, whereas the corresponding full matrix has rank at most $\min\{H_r, d\}$. We therefore apply the calibration

$$\chi_{\text{grp}} = \sqrt{\frac{\min\{H_r, d\}}{G \min\{w, d\}}}, \quad (20)$$

which calibrates the nominal Frobenius scale to that of a full-matrix polar direction. Writing $\mathcal{P}_5$ for this blockwise polar map, Muon’s adjustment is $\mu_{\text{mlp}} = 0.2 \sqrt{\max\{H_r, d\}}$ for either GRAIN matrix role. The parent group direction is $p_{t,i}^r = \mu_{\text{mlp}} \chi_{\text{grp}} \mathcal{P}_5(m_{t,i}^r)$. In parallel, TILLER maintains factorized row and column second moments, in the spirit of memory-efficient factored adaptation (Shazeer & Stern, 2018). For a grouped matrix with gradient g_i and Nesterov momentum m_i , suppressing role and time indices,

$$R_t = \beta_2 R_{t-1} + (1 - \beta_2) \text{rowsum}(g_i^2), \quad (21)$$

$$C_t = \beta_2 C_{t-1} + (1 - \beta_2) \text{colsum}(g_i^2), \quad (22)$$

$$V_{t,jk} = \frac{R_{t,j} C_{t,k}}{\max\{\sum_s R_{t,s}, \delta_{\text{fp}}\}} \frac{1}{1 - \beta_2^t}, \quad a_i = \mu_{\text{mlp}} \frac{m_i}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \quad (23)$$

We set $\beta_2 = 0.95$ and $\epsilon_{\text{opt}} = 10^{-8}$. The same β_2 sets the time scale of the loss ledger in Section D, aligning the adaptive-direction and response histories. The zero-total convention for the factorized estimate is given in Appendix F.3.

B.2 MEASURING RESPONSE BREADTH AND LEARNED CHANGE

The mixing fraction depends on two properties of the activation. Participation measures how broadly its response energy is distributed; congruence compares the learned response with its initialization at the same input. Incoming and outgoing matrices need different measurements: an incoming update acts through J , whereas an outgoing update acts on h . We therefore measure incoming participation from the squared singular values of J , and outgoing participation from the coordinate energies of h .

At a response refresh, let $n \in \{1, \dots, K\}$ index the sampled positions and let $\xi_{n\ell} \in \mathbb{R}^d$ be the downstream cotangent at the output of block ℓ . Define

$$\text{pr}_A(J_{in}) = \frac{[\text{tr}(J_{in} J_{in}^\top)]^2}{w \|J_{in} J_{in}^\top\|_F^2}, \quad \Gamma_{in}^A = J_{in} J_{in}^\top, \quad (24)$$

$$\text{pr}_B(h_{in}) = \frac{\|h_{in}\|_2^4}{w \|h_{in}\|_4^4}, \quad \Gamma_{in}^B = h_{in} h_{in}^\top. \quad (25)$$

For the same normalized input, $\Gamma_{in}^{r,0}$ denotes the kernel obtained from the initial rational response. Let

$$\alpha_{in}^A = \|J_{in}^\top B_i^\top \xi_{n\ell}\|_2^2 / w, \quad \alpha_{in}^B = \|\xi_{n\ell}\|_2^2 / d \quad (26)$$

measure downstream loss sensitivity. Participation and congruence are

$$\pi_i^A = \frac{\sum_n \alpha_{in}^A \text{pr}_A(J_{in})}{\sum_n \alpha_{in}^A}, \quad \pi_i^B = \frac{\sum_n \alpha_{in}^B \text{pr}_B(h_{in})}{\sum_n \alpha_{in}^B}, \quad (27)$$

$$\kappa_i^r = \frac{\sum_n \alpha_{in}^r \langle \Gamma_{in}^r, \Gamma_{in}^{r,0} \rangle}{\sqrt{\left(\sum_n \alpha_{in}^r \|\Gamma_{in}^r\|_F^2 \right) \left(\sum_n \alpha_{in}^r \|\Gamma_{in}^{r,0}\|_F^2 \right)}}. \quad (28)$$

For either role, participation approaches one when energy is uniform and equals $1/w$ when it is concentrated in one singular direction or feature coordinate. The weights α emphasize positions with greater downstream loss sensitivity. Congruence is a cosine between the live and initial kernels, with the same weighting. Thus $1 - (\kappa_i^r)^2$ measures their angular departure. We allocate

$$e_i^r = \pi_i^r (1 - (\kappa_i^r)^2) \quad (29)$$

of the direction energy to an adaptive tangent. This choice gives a large rotation to a response that is both broadly supported and distinct from its initial shape. If the two kernels agree up to a common positive scale, $\kappa = 1$ and the direction remains the Muon parent. Appendix B.4 establishes $e \in [0, 1]$ and specifies zero-response conventions.

B.3 ROTATING THE DIRECTION AT FIXED NORM

Only the component of the adaptive direction orthogonal to the Muon parent can supply a new direction. Removing the parallel component isolates this tangent, and its gradient contraction determines the descent-oriented sign. Suppressing the role index, define

$$a_i^\perp = \begin{cases} a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, & \|p_i\|_F > 0, \\ 0, & \|p_i\|_F = 0, \end{cases} \quad \sigma_i = \begin{cases} 1, & \langle g_i, a_i^\perp \rangle \geq 0, \\ -1, & \langle g_i, a_i^\perp \rangle < 0. \end{cases} \quad (30)$$

When $\|p_i\|_F > 0$ and $\|a_i^\perp\|_F > 0$, form the candidate

$$\tilde{q}_i = \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \quad (31)$$

The selected direction is

$$q_i = \begin{cases} \tilde{q}_i, & \|p_i\|_F \|a_i^\perp\|_F > 0, \langle g_i, p_i \rangle > 0, \langle g_i, \tilde{q}_i \rangle > 0, \text{ and the candidate is finite,} \\ p_i, & \text{otherwise.} \end{cases} \quad (32)$$

Equivalently, the candidate rotates through an angle ϑ_i with $\sin^2 \vartheta_i = e_i$. The factors $\sqrt{1 - e_i}$ and $\sqrt{e_i}$ split the squared norm between two orthogonal components. This is why the response can alter the update direction while leaving its magnitude unchanged.

B.4 BOUNDS AND NORM PRESERVATION

At probe position n , the initialization kernels $\Gamma_{in}^{A,0}$ and $\Gamma_{in}^{B,0}$ evaluate the frozen initial rational coefficients at the current normalized input u_{in} . The input projection and the downstream cotangent remain at their current values. Products between the live and initial Jacobian kernels are evaluated directly from equation 13.

For a nonzero Jacobian with singular values $\{\sigma_j\}_{j=1}^w$,

$$\text{pr}_A(J) = \frac{(\sum_j \sigma_j^2)^2}{w \sum_j \sigma_j^4}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w \|h\|_4^4}. \quad (33)$$

We take $\text{pr}_A(0) = \text{pr}_B(0) = 0$. If either role has zero total sensitivity, we set both participation values to one. If either kernel-alignment denominator vanishes, we set both congruence values to one. In either case equation 29 gives $e_i^r = 0$ whenever the corresponding response comparison is undefined.

Lemma 1 (Response-summary bounds). *For every $J \neq 0$ and $h \neq 0$, $\text{pr}_A(J), \text{pr}_B(h) \in [1/w, 1]$. Whenever the denominator in equation 28 is positive, $\kappa_i^r \in [0, 1]$.*

Proof. For nonnegative v_j , Cauchy–Schwarz gives $(\sum_j v_j)^2 \leq w \sum_j v_j^2$, while $(\sum_j v_j)^2 \geq \sum_j v_j^2$. Apply this first to $v_j = \sigma_j^2$ and then to $v_j = h_j^2$. For congruence, concatenate the weighted kernels as direct-sum matrices $\bigoplus_n \sqrt{\alpha_{in}^r} \Gamma_{in}^r$ and $\bigoplus_n \sqrt{\alpha_{in}^r} \Gamma_{in}^{r,0}$. Frobenius Cauchy–Schwarz gives the upper bound. The numerator is nonnegative because $\langle J J^\top, J^0 J^{0\top} \rangle = \|J^\top J^0\|_F^2$ and $\langle h h^\top, h^0 h^{0\top} \rangle = (h^\top h^0)^2$. $\square$

Proposition 2 (Norm preservation). *Assume $e \in [0, 1]$, $p \neq 0$, and $a^\perp \neq 0$. The candidate in equation 31 satisfies $\|\tilde{q}\|_F = \|p\|_F$, and the selected direction in equation 32 satisfies $\|q\|_F = \|p\|_F$.*

Proof. By construction, $\langle p, a^\perp \rangle = 0$. Since $e \in [0, 1]$,

$$\|\tilde{q}\|_F^2 = (1 - e) \|p\|_F^2 + e \|p\|_F^2 = \|p\|_F^2. \quad (34)$$

The fallback branch in equation 32 is $q = p$. $\square$

C LOSS-COORDINATE MEASUREMENTS

After choosing a direction within each group, TILLER measures how those directions act together. A gradient contraction gives the average descent signal for one group. Shared activation probes add information about where the groups reinforce or offset one another. The resulting score matrix has one column per group direction and one row per sampled position.

C.1 FROM A MATRIX DIRECTION TO A LOSS RESPONSE

On optimizer step 1 and every eight steps thereafter, TILLER deterministically selects a probe set $\mathcal{P}_t$ of $K = 32$ example–token positions. The same positions index every coordinate at that refresh, so each row is a joint measurement across the network. For probe $k \in \mathcal{P}_t$ and coordinate $i = (\ell, g)$, let $x_{k,\ell}$ be the block input, $z_{k,i} = A_i x_{k,\ell}$, $h_{k,i} = \Phi_i(z_{k,i})$. Let $\xi_{k\ell}$ be the corresponding cotangent captured from the mean-reduced backward pass. If a microbatch has R activation positions, M microbatches are accumulated, and χ_{clip} is the realized gradient-clipping factor, define

$$\xi_{k\ell} = RM \chi_{\text{clip}} \bar{\xi}_{k\ell}. \quad (35)$$

This places the cotangent on the summed-loss scale after clipping. Along the paired direction (q_i^A, q_i^B) ,

$$\delta z_{k,i} = q_i^A x_{k,\ell}, \quad \delta h_{k,i} = J_{k,i} \delta z_{k,i}, \quad \delta y_{k,i} = B_i \delta h_{k,i} + q_i^B h_{k,i}. \quad (36)$$

Pairing $\delta y_{k,i}$ with the downstream loss derivative gives

$$S_{t,k,i} = \langle q_i^A x_{k,\ell}, J_{k,i}^\top B_i^\top \xi_{k\ell} \rangle + \langle q_i^B h_{k,i}, \xi_{k\ell} \rangle. \quad (37)$$

At differentiability points, equation 37 is the clipped activation-position contribution to the loss derivative. Averaging over all activation positions recovers the clipped mean-reduced minibatch derivative; sampled rows retain contributions at the probe positions. The exact minibatch contraction is computed below from the parameter gradient. At zero coordinates, the score uses the derivative convention in Appendix A.2. The pullback in equation 13 evaluates its incoming term using groupwise contractions. Applying the same calculation to the decoupled decay direction ($\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i$) and summing over coordinates gives a decay-action vector $a_t^{\text{dec}} \in \mathbb{R}^K$.

For coefficients $c \in \mathbb{R}^C$, the structured directions and decay jointly produce the probe response $S_t c + a_t^{\text{dec}}$. Its squared magnitude is

$$\frac{1}{K} \|S_t c + a_t^{\text{dec}}\|_2^2 = \frac{1}{K} \|S_t c\|_2^2 + \frac{2}{K} c^\top S_t^\top a_t^{\text{dec}} + \frac{1}{K} \|a_t^{\text{dec}}\|_2^2. \quad (38)$$

The last term is constant in c . The first term includes cross products $2c_i c_j \langle S_{t,:i}, S_{t,:j} \rangle / K$ between different groups. Their sign and magnitude determine whether a pair reinforces or offsets across probes. The middle term measures the corresponding interaction with decay. Together they define the quadratic response penalty used to coordinate the directions.

C.2 KEEPING THE MEASUREMENT CURRENT BETWEEN PROBES

Between functional refreshes, the two matrix roles provide the current minibatch directional derivatives

$$s_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle. \quad (39)$$

Let $\nu = \|S_{\text{ref}}\|_F / \sqrt{K}$ be the RMS row norm of the latest functional measurement. Set

$$\gamma_t = \begin{cases} \nu / \|s_t\|_2, & \|s_t\|_2 > 0, \\ 0, & \|s_t\|_2 = 0, \end{cases} \quad S_t = \mathbf{1}_K (\gamma_t s_t)^\top, \quad a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}, \quad (40)$$

where d_t^{dec} is the current minibatch derivative along the complete GRAIN decay direction. This construction exactly preserves the trace of the latest functional geometry because $\|S_t\|_F^2 = K \gamma_t^2 \|s_t\|_2^2 = \|S_{\text{ref}}\|_F^2$ whenever $s_t \neq 0$. Hence every step contributes the current first-order direction through a factor of rank at most one, while functional refreshes periodically restore probe-level cross-coordinate variation.

C.3 DIRECTIONAL SCORE DERIVATION

At one activation position, consider a local intervention with $A_i(\alpha) = A_i + \alpha q_i^A$ and $B_i(\alpha) = B_i + \alpha q_i^B$. At differentiability points, hold the other positions in this block fixed. Differentiating the group contribution $B_i(\alpha) \Phi_i(A_i(\alpha)x)$ at $\alpha = 0$ gives

$$\left. \frac{d}{d\alpha} B_i(\alpha) \Phi_i(A_i(\alpha)x) \right|_{\alpha=0} = q_i^B h_i + B_i J_i q_i^A x, \quad (41)$$

$$\left. \frac{d\mathcal{L}}{d\alpha} \right|_{\alpha=0} = \langle \xi, q_i^B h_i + B_i J_i q_i^A x \rangle. \quad (42)$$

Here $\mathcal{L}$ is the summed microbatch loss multiplied by the realized clipping factor. The clipping factor is held fixed during this differentiation. At a zero coordinate, we use the derivative convention from Appendix A.2. Moving the linear maps across the inner product yields equation 37. Thus one row of S records the loss-sensitive contributions of every layer-group direction at the same sampled position; its rowwise outer product couples coordinates whose contributions vary together.

The indices used to sample x , z , h , and $\bar{\xi}$ are identical within each probe row.

D A ROBUST FREQUENT-DIRECTIONS LOSS LEDGER

The loss ledger summarizes recent interactions between group directions. Its target is the discounted second moment of the score rows,

$$G_t^* = \beta_2 G_{t-1}^* + (1 - \beta_2) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K. \quad (43)$$

For a coefficient vector c , $c^\top G_t^* c$ measures the recent squared loss response of that combination. Keeping the off-diagonal information lets the optimizer distinguish combinations with similar individual descent signals but different joint responses. Each historical score row is evaluated using the parameters and directions at the step when that row is recorded.

D.1 RETAINING DOMINANT RESPONSES AND A SPECTRAL REMAINDER

TILLER represents the second moment by a low-rank factor and an isotropic remainder. The factor retains dominant combinations of response coordinates; the remainder adds an equal penalty in all coefficient directions, including those outside the retained subspace. It initializes $\mathcal{B}_1 = S_1$ and $\tau_1 = 0$. From the second step onward, it retains $\mathcal{B}_{t-1}$ with at most $r = 64$ rows and forms

$$\mathcal{A}_t = \begin{bmatrix} \sqrt{\beta_2} \mathcal{B}_{t-1} \\ \sqrt{1 - \beta_2} S_t \end{bmatrix}. \quad (44)$$

Following Frequent Directions and its robust isotropic completion (Ghashami et al., 2016; Luo et al., 2019), TILLER diagonalizes the row Gram matrix $\mathcal{A}_t \mathcal{A}_t^\top$. If Δ_t is the largest discarded eigenvalue, it shrinks every retained eigenvalue by Δ_t and adds $\Delta_t/(2K)$ to an isotropic tail. The persistent metric is

$$\hat{G}_t = \frac{1}{K} \mathcal{B}_t^\top \mathcal{B}_t + \tau_t I, \quad \tau_t = \beta_2 \tau_{t-1} + \frac{\Delta_t}{2K}. \quad (45)$$

The midpoint tail assigns half of the unrepresented eigenvalue interval to every direction. It reduces the worst-case completion bound over $[0, \Delta_t/K]$ from Δ_t/K to $\Delta_t/(2K)$, yielding the bound in Appendix D.2. The score-decay cross term is

$$r_t = \beta_2 r_{t-1} + (1 - \beta_2) S_t^\top a_t^{\text{dec}} / K. \quad (46)$$

The current coefficient proposal uses the old compressed history together with all of the new rows, before their compression:

$$G_t = \mathcal{A}_t^\top \mathcal{A}_t / K + \text{Diag}(\delta_t), \quad \delta_t = \beta_2 \tau_{t-1} \mathbf{1}. \quad (47)$$

This is the G_t in equation 6. The compressed state $(\mathcal{B}_t, \tau_t, r_t)$ is carried to the next step. At initialization, $\mathcal{A}_1 = S_1$, $\delta_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$.

D.2 WHY THE MIDPOINT REMAINDER CONTROLS APPROXIMATION ERROR

Shrinking the retained spectrum leaves an omitted contribution between zero and Δ/K in every direction. Assigning its midpoint to the isotropic remainder limits the error to half that interval. The following result makes this statement precise for the metric used in the coefficient model.

Proposition 3 (Positive semidefiniteness and spectral error). *Let $K > 0$, $0 \leq \beta_2 \leq 1$, $\tau_{t-1} \geq 0$, and $\mathcal{A} = U \text{Diag}(\sigma) V^\top$, with singular values in nonincreasing order. If $\text{rank}(\mathcal{A}) > r$, set $\Delta = \sigma_{r+1}^2$ and define*

$$\mathcal{B} = \text{Diag}\left(\sqrt{(\sigma_1^2 - \Delta)_+}, \dots, \sqrt{(\sigma_r^2 - \Delta)_+}\right) V_{1:r}^\top. \quad (48)$$

If $\text{rank}(\mathcal{A}) \leq r$, set $\Delta = 0$ and choose any factor with at most r rows satisfying $\mathcal{B}^\top \mathcal{B} = \mathcal{A}^\top \mathcal{A}$. Let

$$G = \mathcal{A}^\top \mathcal{A} / K + \beta_2 \tau_{t-1} I \quad (49)$$

be the pre-compression metric, and form $\hat{G} = \mathcal{B}^\top \mathcal{B} / K + (\beta_2 \tau_{t-1} + \Delta/(2K))I$. Then

$$\hat{G} \succeq 0, \quad \|\hat{G} - G\|_2 \leq \frac{\Delta}{2K}. \quad (50)$$

Proof. When $\text{rank}(\mathcal{A}) \leq r$, the two metrics are identical and the claim follows with $\Delta = 0$. Otherwise, the retained eigenvalues after shrinkage are nonnegative, and $\tau_t = \beta_2 \tau_{t-1} + \Delta/(2K) \geq 0$, proving positive semidefiniteness. On a retained right-singular direction, the represented eigenvalue differs from the live eigenvalue by $-\Delta/(2K)$. On a discarded direction with live eigenvalue $\lambda \in [0, \Delta]$, the error is $(\Delta/2 - \lambda)/K$. On a null direction it is $\Delta/(2K)$. The largest absolute eigenvalue error is therefore at most $\Delta/(2K)$. $\square$

This bound controls the quadratic response penalty for every proposed combination of directions. For any c , spectral-norm Cauchy–Schwarz gives

$$|c^\top \widehat{G}c - c^\top Gc| \leq \|c\|_2 \|(\widehat{G} - G)c\|_2 \leq \frac{\Delta}{2K} \|c\|_2^2. \quad (51)$$

Thus the discarded spectral threshold determines how much compression can change the response penalty, including for signed coefficient vectors.

The midpoint minimizes the worst absolute scalar-completion error over $[0, \Delta/K]$. The one-step result also yields a discounted history bound. For the uncompressed target in equation 43, if $E_t = \widehat{G}_t - G_t^*$, then

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta_2^{t-j} \frac{\Delta_j}{2K}. \quad (52)$$

Indeed, the compression error at step t is a perturbation D_t with $\|D_t\|_2 \leq \Delta_t/(2K)$, so $E_t = \beta_2 E_{t-1} + D_t$ and $E_1 = 0$.

E COORDINATING GROUP DIRECTIONS AT FIXED TOTAL NORM

The response-aware directions specify how to update each group. Coordination chooses their relative coefficients using the current descent signal and the loss ledger. Fixing the total directional norm makes this a choice of how to distribute the update across groups. Signed coefficients allow a group to reverse direction when the joint proposal improves the model and preserves descent for both matrix roles in each layer.

E.1 THE COEFFICIENT MODEL

Let $c \in \mathbb{R}^C$ scale the paired directions and define $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$. Assume $\omega_i > 0$ for every coordinate. The uncoordinated update is $c = \mathbf{1}$, whose squared norm is $\mathcal{E}_0 = \sum_i \omega_i$. Indeed, the parameter blocks associated with different coordinates are disjoint, so the structured update $\Delta\theta(c) = -\eta\{c_i q_i^A, c_i q_i^B\}_{i=1}^C$ satisfies $\|\Delta\theta(c)\|_2^2 = \eta^2 \sum_i \omega_i c_i^2$. With current directional derivatives s_t , pre-compression factor $\mathcal{A}_t$, diagonal completion δ_t , score–decay cross term r_t , and scheduled step size η , TILLER forms the response penalty

$$\mathcal{R}_t(c) = \frac{1}{K} \|\mathcal{A}_t c\|_2^2 + \delta_t^\top (c \odot c) + 2r_t^\top c \quad (53)$$

and forms the constrained model

$$\min_c F_t(c) = -\eta s_t^\top c + \frac{\eta^2}{2} \mathcal{R}_t(c), \quad (54)$$

$$\text{s.t. } \sum_i \omega_i c_i^2 = \mathcal{E}_0. \quad (55)$$

The linear term rewards descent on the current minibatch. The quadratic term measures the joint squared response of the group directions, including their interaction with decay. Its cross terms can increase or decrease the penalty as coefficients change together. The constraint gives the coordinated GRAIN direction the same aggregate squared norm as the uncoordinated direction, before their common decay term is applied.

After whitening $y = W^{1/2}c$, where $W = \text{Diag}(\omega)$, the stationarity equation is

$$(M_t + \lambda I)y = b_t, \quad M_t = W^{-1/2} (\mathcal{A}_t^\top \mathcal{A}_t / K + \text{Diag}(\delta_t)) W^{-1/2}, \quad b_t = W^{-1/2} (s_t / \eta - r_t). \quad (56)$$

The solver below proposes a coefficient vector. We retain it when it satisfies the norm constraint, lowers F_t relative to $c = \mathbf{1}$, and preserves positive layerwise gradient and momentum contractions

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle > 0 \quad (r \in \{A, B\}). \quad (57)$$

Both contractions must also be positive for the uncoordinated coefficients $c = \mathbf{1}$. If a proposal fails these checks, the update uses $c = \mathbf{1}$.

E.2 PROJECTED SOLUTION

We solve for a proposal in a subspace generated by the current right-hand side and the response metric. Whitening converts the weighted norm constraint to a sphere; projection then reduces the multiplier search to a scalar equation. The final coefficients are evaluated in the original model, so acceptance depends on the full response penalty rather than its projected approximation.

For $\eta > 0$, $\omega_i > 0$, and $\delta_t \geq 0$, let $W = \text{Diag}(\omega)$ and set $y = W^{1/2}c$. Substituting $c = W^{-1/2}y$ into equations 54 and 55, then dividing by η^2 , gives the equivalent problem

$$\min_{\|y\|_2^2 = \mathcal{E}_0} \frac{1}{2} y^\top M_t y - b_t^\top y, \quad (58)$$

with M_t and b_t from equation 56. Its Lagrangian

$$\mathcal{J}(y, \lambda) = \frac{1}{2} y^\top M_t y - b_t^\top y + \frac{\lambda}{2} (\|y\|_2^2 - \mathcal{E}_0) \quad (59)$$

has stationarity equation $(M_t + \lambda I)y = b_t$.

For $b_t \neq 0$, matrix products are evaluated as $M_t v = W^{-1/2} \mathcal{A}_t^\top (\mathcal{A}_t W^{-1/2} v) / K + (\delta_t \odot \omega) \odot v$. A Lanczos recurrence (Lanczos, 1950) constructs an orthonormal basis Q_k , with $k \leq 32$. Without breakdown, this basis spans $\mathcal{K}_k(M_t, b_t) = \text{span}\{b_t, M_t b_t, \dots, M_t^{k-1} b_t\}$. Let T_k^+ be the positive-semidefinite projection of the accumulated projected matrix. The solver uses

$$(T_k^+ + \lambda I)z = Q_k^\top b_t, \quad \|z\|_2^2 = \mathcal{E}_0. \quad (60)$$

With $\gamma = U^\top Q_k^\top b_t$, the norm constraint becomes

$$\psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (61)$$

On the regular branch $\lambda > 0$, ψ is continuous and strictly decreasing because $\gamma \neq 0$ and $\theta_j \geq 0$. A positive root, when it exists in the implemented interval, is obtained by bisection. Otherwise the uncoordinated coefficients are retained. Appendix F.3 specifies the restart, projection, and root bracket.

The projected coefficient vector is evaluated in double precision using the unprojected low-rank-plus-diagonal objective. A finite proposal is retained when

$$\frac{|c^\top W c - \mathcal{E}_0|}{\max\{\mathcal{E}_0, 1\}} \leq 10^{-8}, \quad F_t(c) < F_t(\mathbf{1}), \quad (62)$$

and the aggregate current-gradient and Nesterov contractions are positive for both matrix roles in each layer, for both the proposal and the uncoordinated update.

F COMPLETING THE TRANSFORMER UPDATE

Attention uses the response information at the layer level. Aggregating over GRAIN groups produces one rotation fraction for the attention matrices in that layer. The groupwise coefficient model remains attached to the paired GRAIN projections, whose loss responses were measured above.

F.1 LAYERWISE RESPONSE GUIDANCE FOR ATTENTION

For attention, the group statistics are summarized at each layer as

$$\pi_\ell^{\text{att}} = \sqrt{\left(G^{-1} \sum_g \pi_{\ell g}^A\right) \left(G^{-1} \sum_g \pi_{\ell g}^B\right)}, \quad \kappa_\ell^{\text{att}} = \sqrt{\kappa_\ell^A \kappa_\ell^B}, \quad (63)$$

where each κ_ℓ^r is formed by summing the numerator and two denominator terms of equation 28 over groups before normalization. With

$$e_\ell^{\text{att}} = \pi_\ell^{\text{att}} [1 - (\kappa_\ell^{\text{att}})^2], \quad (64)$$

the participation factor is largest when both matrix roles have broad support, while the departure factor opens when either role has moved away from its initial response. These layerwise statistics gate the same norm-preserving chord for the QKV and output matrices. The 16 attention heads are partitioned into blocks of four for the polar map; the stacked QKV direction is scaled by $1/\sqrt{3}$ to match its three constituent projections. The adaptive tangent is factorized at matrix level. Attention does not enter the LG coefficient model. Learned rational coefficients, embeddings, normalization parameters, and remaining tensors use AdamW (Loshchilov & Hutter, 2019).

F.2 COMPLETE UPDATE

For $C \geq 2$, $\omega_i > 0$, and $b_t \neq 0$, one step updates the momentum, factorized moments, response summaries, and loss ledger as follows.

1. Update momentum and construct the five-step Newton–Schulz Muon parent directions for the paired GRAIN matrices and the attention matrices.
2. Update the factorized second moments. On a functional refresh, compute participation and congruence; between refreshes, reuse their latest values. Form the response-gated tangent chords.
3. At step 1 and every eight steps thereafter, evaluate the 32 rows in equation 37 and the corresponding decay action. At other steps, form the trace-matched rows from the current batch-gradient scores.
4. Form the current pre-compression factor $\mathcal{A}_t$ using the old factor and the current rows, then update the stored at-most-rank-64 Robust Frequent-Directions factor, isotropic tail, and score–decay cross term.
5. Using the pre-compression factor and the old isotropic tail, construct the at-most 32-dimensional Lanczos basis, attempt the positive regular branch of equation 61, and evaluate the coefficient proposal.
6. Scale both GRAIN matrix directions in coordinate (ℓ, g) by $c_{\ell g}$, then apply decoupled decay to the existing parameters and add the scheduled directional update. Apply the response-informed chord to the attention matrices.
7. Update the rational coefficients and remaining parameter classes with AdamW.

F.3 NUMERICAL SPECIFICATION

The five-step polar map starts from the input matrix normalized by its Frobenius norm (lower bounded by 10^{-7}), transposing tall matrices before the iteration. The normalization norm is evaluated in float32 and the polar iteration runs in bfloat16. Each step is

$$X_{j+1} = aX_j + b(X_j X_j^\top)X_j + c(X_j X_j^\top)^2 X_j, \quad (a, b, c) = (3.4445, -4.775, 2.0315), \quad (65)$$

after which the original orientation is restored. The Lanczos recurrence is fully reorthogonalized. If its breakdown threshold is reached early, a deterministic cosine vector is reorthogonalized against the existing basis to start a new block, with zero coupling between blocks; the basis is used only after a final orthogonality check. If $T_k = U \text{Diag}(\bar{\theta}) U^\top$ is the accumulated projected matrix, the solver uses $T_k^+ = U \text{Diag}((\bar{\theta})_+) U^\top$. Its positive multiplier is bracketed by

$$\lambda_{\text{lo}} = \epsilon_{\text{mach}} C, \quad \lambda_{\text{hi}} = \frac{\|b_t\|_2}{\sqrt{\mathcal{E}_0}} + \lambda_{\text{max}}(T_k^+) + \lambda_{\text{lo}}, \quad (66)$$

and approximated by 64 bisection steps when $\psi(\lambda_{\text{lo}}) \geq \mathcal{E}_0$. Otherwise the coefficient proposal is rejected. The factorized moments use float32 accumulators and $\delta_{\text{fp}} = \text{finfo}(\text{float32}).\text{tiny}$; when the row-moment total is zero, the estimate in equation 23 is $V_t = 0$.